

Name: _____

Class: _____

Index: _____

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

BIOLOGY
Paper 2 Core Paper

9747/02
Wednesday 27 August 2008

2 hours

Additional materials: Answer booklet
Additional Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A Structured Questions

Answer **all** questions.
Write your answers in the spaces provided on the question paper.
Show All working

Section B Essay Questions

Answer any **one** questions.
Write your answers on the separate answer paper.
All working for numerical answers must be shown.

At the end of the examination,

1. fasten all your work securely together;
2. enter the number of the Section B question you have answered in the grid opposite.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	80
1 [11]	
2 [13]	
3 [16]	
4 [10]	
5 [17]	
6 [13]	
Section B	20
7 [20]	
8 [20]	
Total	100

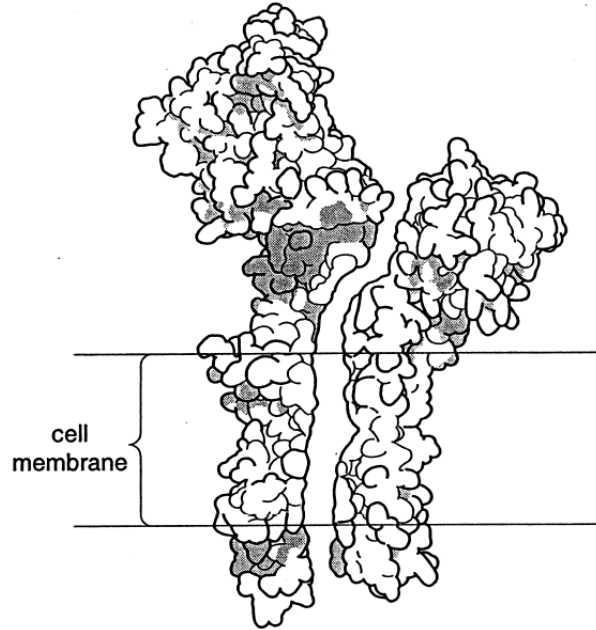
This document consists of **14** printed pages.

[Turn over]

Section A

- 1 Fig. 1.1 shows the molecular structure of a protein pump that moves calcium ions across cell membranes.

Fig. 1.1



- (a) Explain why such pumps are often necessary to move calcium ions across membranes. [3]

- (b) Outline how such a protein pump would move calcium ions across the membrane shown. [2]

Transmission of an impulse across a synapse is triggered by a sudden influx of calcium ions into the synaptic knob. This does not occur through a protein pump.

- (c) Describe how this influx of calcium ions occurs through the presynaptic membrane. [2]

- (d) Explain how the entry of calcium ions into the synaptic knob leads to the impulse being transmitted across the synapse. [4]

[Total: 11]

- 2 Many bryophytes grow in shady conditions in woodlands and forests where the lack of direct sunlight and the cool, moist conditions reduce the risk of desiccation. However, some species do survive in open, sunny habitats.

The figure below shows how the rate of photosynthesis varies with light intensity, for two bryophytes, *Jubula hutchinsiae* grows in forests, while *Tortula ruralis* grows in sunny, exposed habitats.

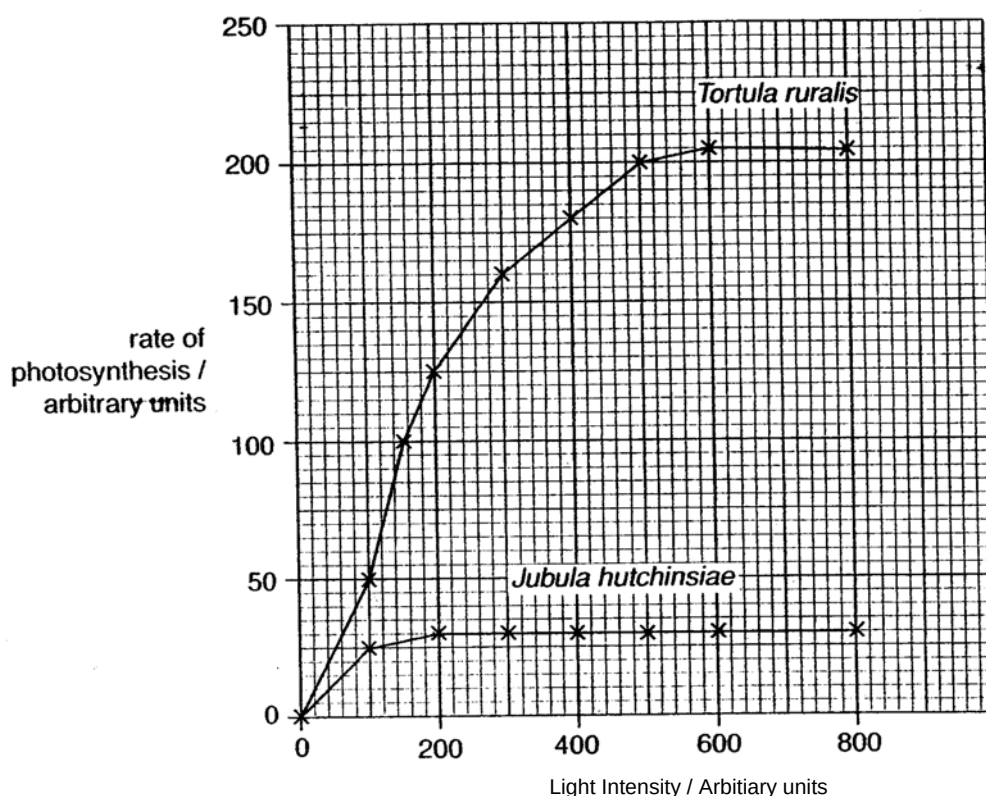


Fig 2.1

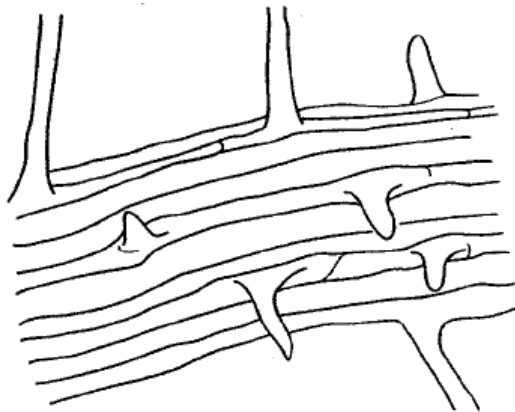
- (a) (i) With reference to Fig 2.1, describe how light intensity affects the rate of photosynthesis of *T. ruralis*. [2]

(ii) explain the effect you have described in (2ai)

[2]

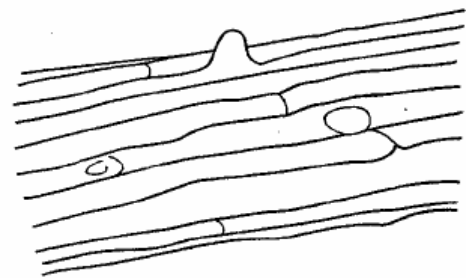
(iii) Suggest with reasons, why the rate of photosynthesis of *J. hutchinsiae* reaches a plateau at a much lower light intensity than that of *T. ruralis*. [2]

(b) In another species of plant called *Arabidopsis thaliana* also found in the forested areas, a mutation of a gene coding for the enzyme reduced NADP oxidase results in plants with very very short root hair. The root hair of wild type and mutant plants are shown in Fig 2.2.



Wild Type Plant

Fig 2.2



Mutant Plant

Reduced NADP oxidase converts reduced NADP to NADP. Explain the importance of reduced NADP in the process of photosynthesis. [2]

The length of root hair in wild and mutant, *A. thaliana* plants were measured at intervals, together with the activity of reduced NADP oxidase in the tips of the root hairs. The results are shown in Fig 2.3

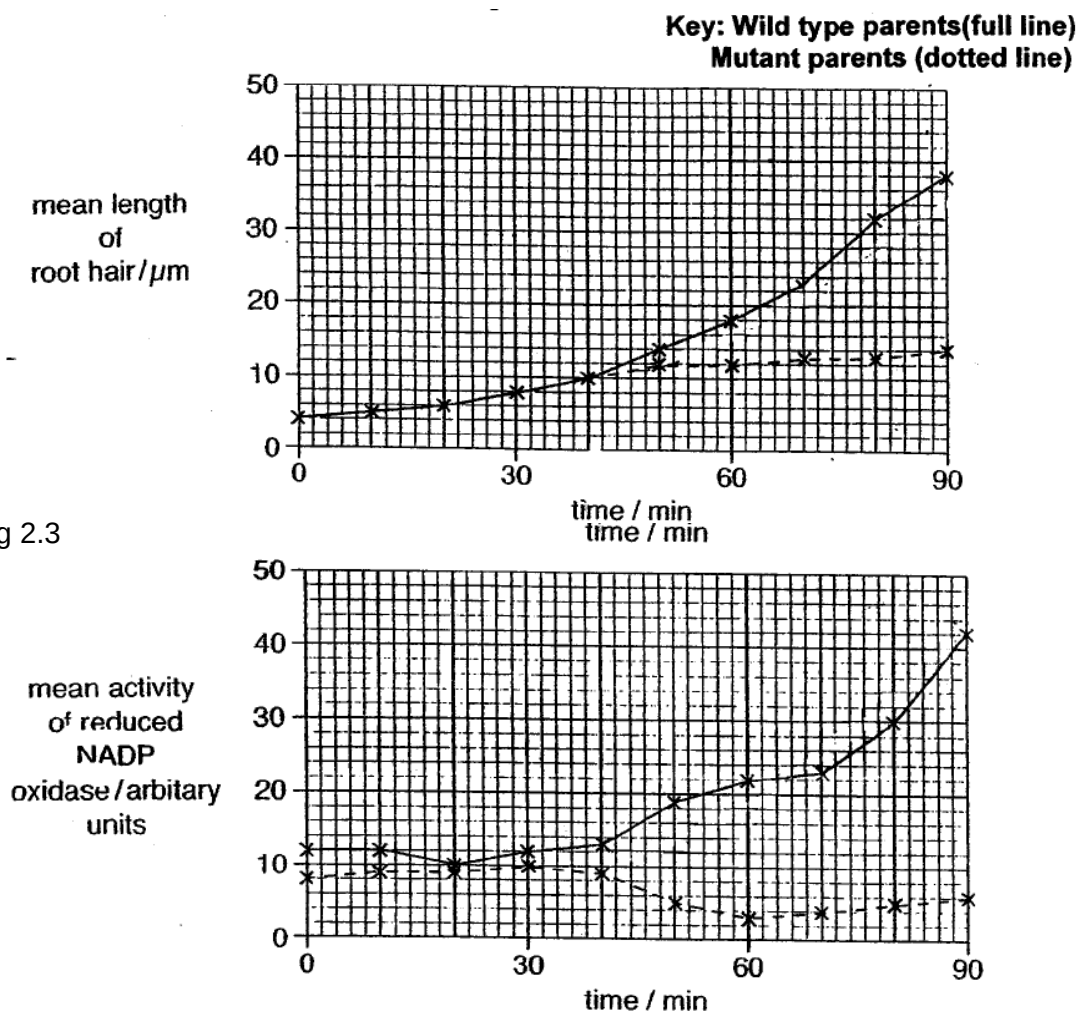
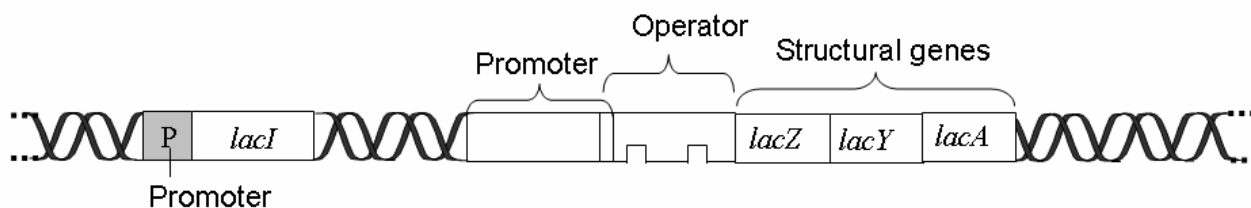


Fig 2.3

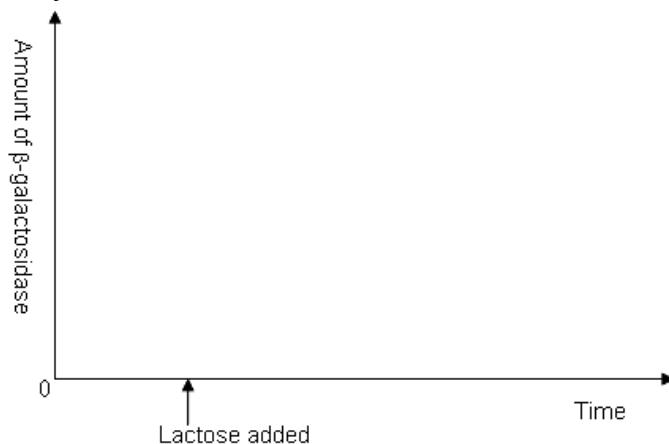
- (c) With reference to Fig 2.3 above, suggest an explanation, making use of your knowledge and understanding of photosynthesis, for the difference in the growth of root hair in the two types of plant. [5]

[Total: 13]

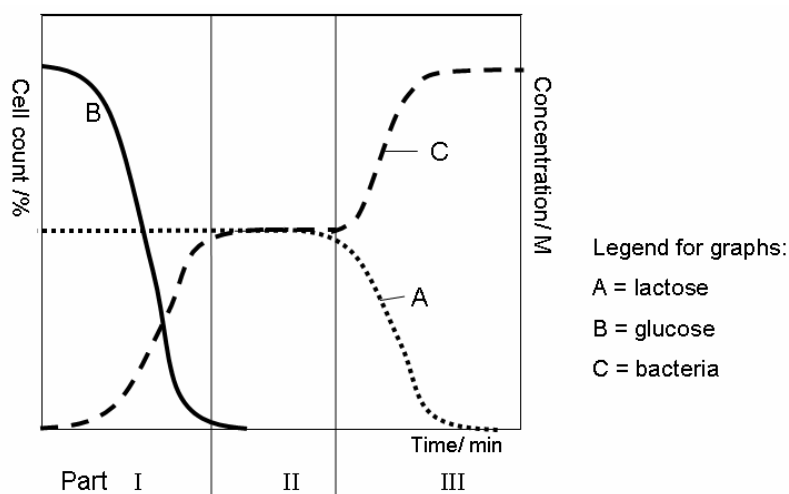
- 3 A study of the bacterium's chromosome showed that it possessed a *lac* operon to allow the bacterium to metabolise lactose. The structure of the *lac* operon is shown below:



- (a) On the axes below, draw a graph showing how the amount of β -galactosidase in the bacterium is affected by the addition of lactose. [2]



- (b) The graph below shows the growth of bacteria in the presence of glucose and lactose.



Describe and explain the three parts of graph C from the view point of the *lac* operon.

Part I:

[6]

Part II:

[3]

Part III:

[5]

[Total:16]

- 4 A series of breeding experiments were carried out with *Drosophila*, the fruit fly.

A cross between pure breeding males with red eyes and straight wings and females with purple eyes and curled wings produced offspring all with red eyes and straight wings. The same results are obtained when the parents of the same pure-breeding phenotypes were used again but with the sexes reversed.

- (a) Suggest a reason for carrying out the cross with the sexes reversed. [1]

Female offspring from the first experiment were then crossed with purple eyed curled winged males.

- (b) Draw a genetic diagram to show the results of this cross. [4]

A null hypothesis was put forward that the genes for eye colour and wing shape are not linked. The chi-squared (χ^2) test was applied to the data obtained. The formula is given below:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- (c) (i) Complete the table and use the formula to determine the value of χ^2 for the results of this investigation. Show working. [2]

Phenotype	red eye, straight wing	red eye, curled wing	purple eye straight wing	purple eye, curled wing
Observed				
Expected				

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- (ii) Use the table to determine if this hypothesis was correct. Explain how you arrive at your answer. [3]

Degrees of freedom	Probability value					
	0.99	0.95	0.10	0.05	0.01	0.001
1	0.0002	0.0039	2.71	3.84	6.63	10.83
2	0.020	0.103	4.61	5.99	9.21	13.82
3	0.115	0.352	6.25	7.81	11.34	16.27
4	0.297	0.711	7.78	9.49	13.28	18.47

Degree of Freedom:

χ^2 tabulated :

Assumptions:

[Total : 10]

- 5 In most cases of chronic myeloid leukemia, the cells share a chromosome abnormality not found in other non-leukemic white blood cells, or any other cells in the body. This abnormality is shown in Fig. 5.1 below.

The Philadelphia chromosome (Ph) is an abnormally short chromosome 22 that is one of the two chromosomes involved in genetic exchange with chromosome 9. It is the first consistent abnormality found in any kind of malignancy.

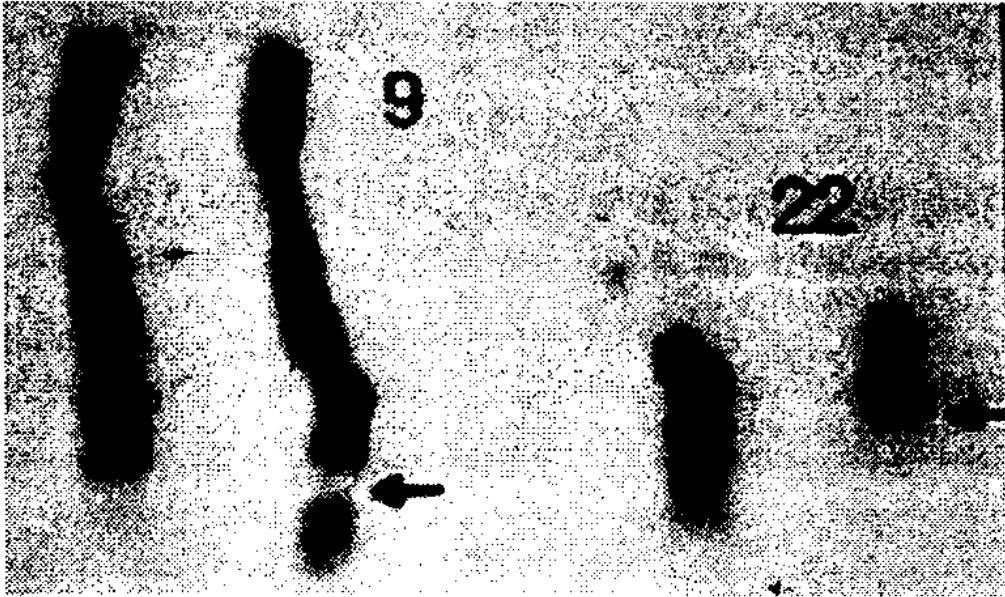


Fig. 5.1

- (a) Deduce the kind of genetic exchange that occurs between chromosome 9 and 22 in Fig 5.1. [1]

The DNA removed from chromosome 9 contains most of a gene titled **c-ABL**. c-ABL produces an ABL protein that acts as a tyrosine kinase. The break in chromosome 22 occurs in the middle of a gene titled **BCR**. The resultant Ph chromosome has the BCR fused to most of c-ABL.

Transcription and translation of the hybrid BCR-ABL gene produces an abnormal 'fusion' protein that activates **constitutively** (i.e. all the time) a number of cell activities that are normally turned on only in the presence of a growth factor, such as a Platelet Derived Growth Factor (PDGF). The fusion protein also functions as a tyrosine kinase. This mutation occurs in a single bone marrow cell, and by mitosis, the number of Ph chromosome containing cells increases.

- (b) (i) Based on the description of the fusion protein, what can one deduce about the type of gene the c-ABL belongs to? Explain your answer. [1]

-
- (ii) Predict two effects this hybrid protein has on the cell. [1]
-

During the chronic phase of this disease, these cells are still able to exit the cell cycle and perform their normal function.

- (c) Using your knowledge of cell signaling, describe the sequence of events that occurs to a cell for it to become differentiated. [3]

At some point another mutation in a tumour suppressor gene or a proto-oncogene will occur in one of these cells. This additional mutation causes the rate of mitosis in these cells to increase sharply. The daughter cells fail to differentiate and a patient enters the crisis phase of the disease.

Presently, a drug named Gleevec[®] is being used to treat patients with CML. Patients given Gleevec[®] have so far shown return to normal blood counts and disappearance of leukemia cells in the marrow. Fig. 5.2 shows how Gleevec works.

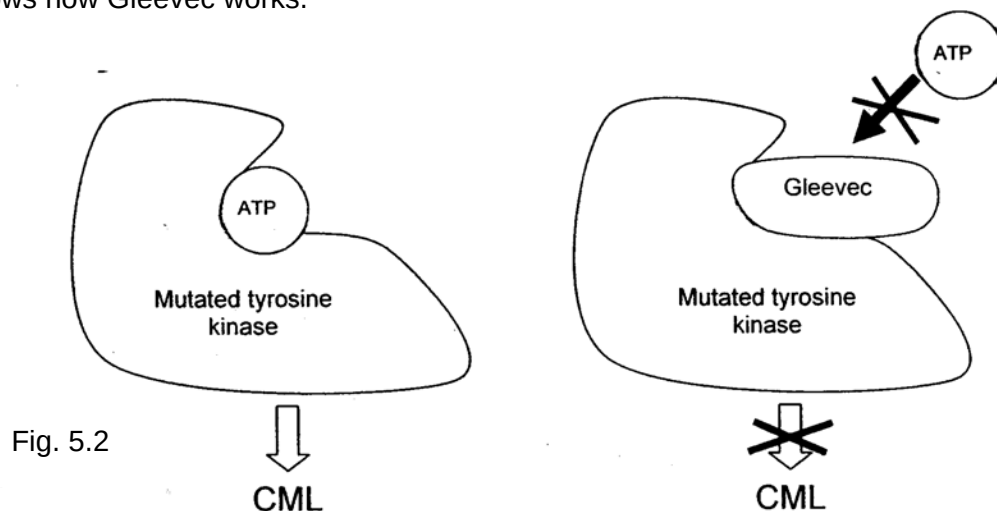


Fig. 5.2

- (d) Describe the role of tumour suppressor genes in normal cells. [2]

- (e) Predict whether a mutation in the tumour suppressor gene will be a dominant or recessive mutation. Give reasons for your answer. [2]

(f) Name one other way in which a cell's ability to proliferate can become uncontrolled. [1]

(g) Briefly explain the mode of action of Gleevec in controlling CML. [2]

Recently two different point mutations have been detected in the ATP binding site which affects the binding of Gleevec, leading to resistance. Table 5.3 summarises the two types of mutations that have been detected.

Table 5.3 Showing the types of mutations to ABL-BCR and their effects

<i>Types</i>	<i>Mutation 1</i>	<i>Mutation 2</i>
Change of base	thymine to cytosine	cytosine to thymine
Amino acid change	methionine to threonine	threonine to isoleucine
Resistance to Gleevec	partial	complete
Recommended treatment	higher dosage of Gleevec	another drug

(h) How does a single base change lead to resistance to Gleevec? [2]

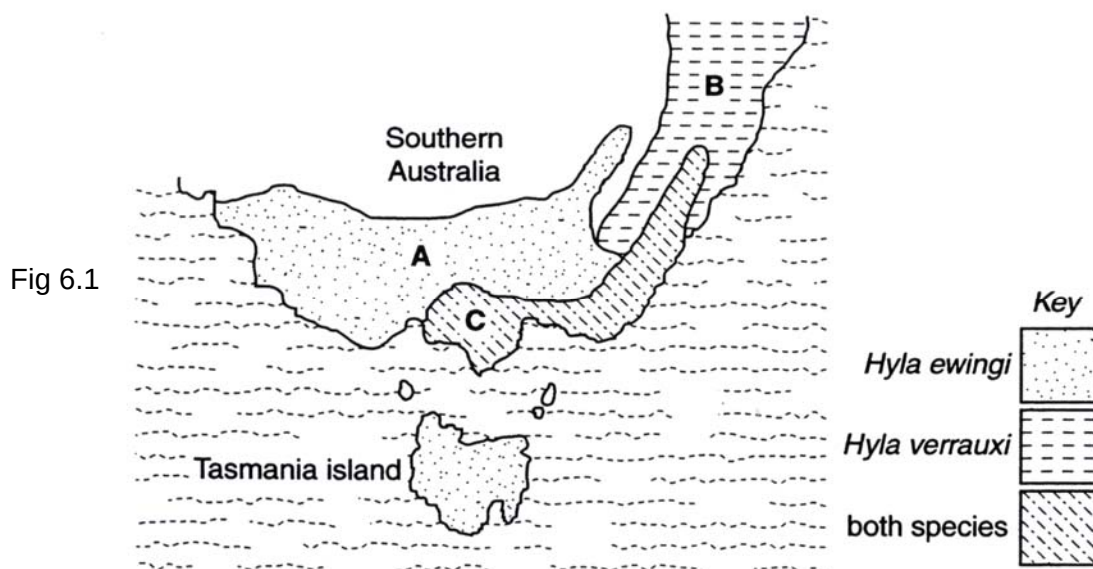
(i) Suggest possible reasons for the differences in the resistance to Gleevec and the recommended treatment despite both mutations involving one amino acid change. [2]

[Total: 17]

- 6 (a) Two closely related species of frog, *Hyla ewingi* and *Hyla verrauxi* live in South Australia. See Fig 6.1
Explain what is meant by the terms;
(i) *Niche* [2]

- (ii) *Species* [3]

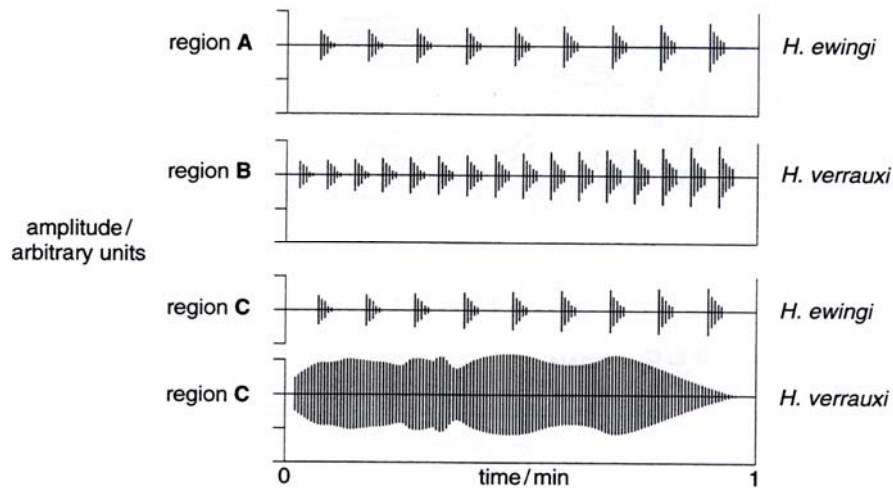
The figure below shows the distribution of the tree frogs in Southern Australia.



Hyla ewingi and *Hyla verrauxi* are two closely related species of tree frogs from southern Australia. Research from breeding studies and DNA sequence data has shown that they have weak genetic incompatibility.

Male frogs attract females of the same species for mating by their pulsing call. The pulse rate of the male calls of the two species is almost identical. However, when both species coexist within the same region, the calls of *H. ewingi* are quite different than those of *H. verrauxi*.

The following data shows the calls of *H. ewingi* and *H. verreauxi* from three regions, A, B and C. The male advertisement calls were recorded and played back electronically to produce the pattern shown



Some female frogs of the species *H. verrauxi* were transferred from region B to region C. Observations shown that they were not attracted by the calls of the males in region C.

- (b)** Suggest why this is so. [3]

- (c) Explain how the process of natural selection could have led to the different male advertisement call in *H. verreauxi* in region C. [5]

[illegible]

[Total: 13]

Section B

Answer **ONE** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 7 (a) Describe and explain the secondary structure of proteins? [6]
 (b) Describe the main ways in which a globular protein differs from DNA. [8]
 (c) Using HIV protease as an example, analyze how its structure is related to its function. [6]
[Total:20]

or

- 8 (a) Describe the structure and function of the parts of a eukaryotic chromosome. You may wish to include a diagram as part of your description. [10]
 (b) Explain how prokaryotic genome structure differs from eukaryotes. [6]
 (c) Discuss the evolutionary significance of organizing genes into chromosomes. [4]
[Total:20]